

SHAWANO, WI, USA

WMO: 720589

Lat: 44.783N

Lon: 88.550W

Elev: 814

StdP: 14.27

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00185

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -2.1	(d) -16.8	(e) 2.3	(f) -5.3	(g) -12.2	(h) 2.9	(i) -1.1	(j) 19.2	(k) 21.4	(l) 17.5	(m) 22.4	(n) 3.1	(o) 270	(p) 0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.8	125.5	80.5	71.3	118.8	78.8	69.5	111.7	77.5	39.5	84.9	37.8	82.0	36.2	80.3	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	16.2	13.6	-13.2	93.8	3.9	2.5	-15.9	95.6	-18.2	97.1	-20.4	98.5	-23.2	100.3
			-13.6	78.5	3.8	2.6	-16.4	80.3	-18.7	81.9	-20.8	83.4	-23.6	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	19.4	21.8	33.3	44.5	58.6	67.3	71.8	69.6	61.9	49.7	36.8	25.7
	DBStd	20.39	11.71	10.88	11.34	8.73	8.74	6.32	5.39	5.30	7.57	7.97	9.41	10.32
	HDD50	3770	950	790	529	209	21	0	0	0	5	108	404	754
	HDD65	7222	1415	1209	983	618	238	46	8	16	146	479	845	1219
	CDD50	2613	0	0	12	42	288	518	676	609	362	97	9	0
	CDD65	589	0	0	0	1	39	114	220	160	52	3	0	0
	CDH74	5122	0	0	10	29	465	973	1991	1191	425	38	0	0
	CDH80	1442	0	0	0	5	140	283	638	291	83	2	0	0
	WSAvg	5.5	5.5	5.9	5.8	7.1	6.5	5.4	4.4	4.0	4.6	5.6	5.3	5.4
	PrecAvg	31.5	1.2	0.9	1.7	2.8	3.8	4.2	3.9	3.5	3.4	2.7	1.9	1.5
(15) Precipitation	PrecMax	37.0	3.5	2.0	3.3	4.9	5.7	8.0	8.8	9.0	7.6	5.7	4.9	3.0
	PrecMin	26.6	0.3	0.2	0.2	0.6	1.7	1.8	1.1	0.7	0.8	0.2	0.2	0.2
	PrecStd	3.3	0.7	0.4	0.8	1.1	1.2	1.9	1.9	1.7	1.8	1.5	1.2	0.7
	PrecStd	3.3	0.7	0.4	0.8	1.1	1.2	1.9	1.9	1.7	1.8	1.5	1.2	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.5	52.8	76.6	78.1	89.8	91.3	92.9	89.8	87.3	79.0	67.6	54.0
		MCWB	39.7	45.7	62.1	60.3	68.7	73.2	75.9	74.1	72.7	63.4	56.6	50.1
	2%	DB	40.6	45.4	63.6	70.4	83.9	86.7	88.7	86.1	82.2	72.8	59.5	45.2
		MCWB	36.9	40.8	53.2	56.1	65.1	69.5	73.6	72.3	69.4	60.3	50.4	41.6
	5%	DB	37.3	40.2	55.3	64.9	79.3	83.2	86.3	83.3	79.2	68.0	54.6	41.4
		MCWB	34.8	35.5	45.7	51.7	63.3	68.4	71.6	70.8	66.7	59.2	48.3	38.5
	10%	DB	35.2	36.6	49.6	60.5	74.6	79.8	83.7	81.0	75.5	63.2	50.2	38.3
		MCWB	33.2	32.9	42.5	48.4	61.1	66.0	69.9	68.5	66.4	55.8	45.4	35.9
	0.4%	WB	40.2	46.4	63.0	64.5	72.0	76.7	79.6	76.7	74.4	68.4	57.5	50.9
		MCDB	44.1	51.2	76.7	73.6	83.7	87.4	90.6	84.6	82.3	73.2	64.8	52.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.6	40.0	55.1	58.4	69.0	73.2	76.3	74.6	72.2	63.4	52.5	42.4
		MCDB	40.4	44.9	60.9	67.6	78.4	82.7	84.8	82.8	79.6	70.0	57.6	44.0
	5%	WB	35.2	36.2	47.3	53.3	66.3	71.0	74.2	72.8	70.0	60.1	48.8	39.0
		MCDB	37.0	39.6	55.5	63.4	75.3	78.6	82.3	80.4	76.4	67.0	53.7	41.5
	10%	WB	33.2	33.2	42.4	49.3	63.2	68.4	72.2	71.0	67.8	56.7	45.8	36.2
		MCDB	35.0	36.2	48.6	58.6	71.9	76.4	80.4	78.1	73.4	62.1	49.4	38.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	16.9	18.8	19.1	21.4	19.5	20.3	19.1	20.2	17.0	14.4	12.0
		MCDBR	14.5	18.4	25.7	28.7	27.8	23.2	22.7	20.9	24.0	25.0	22.4	14.5
		MCWBR	12.0	13.5	16.3	16.6	12.8	11.0	10.7	9.3	12.0	14.7	14.5	11.9
	5% WB	MCDBR	13.9	16.9	24.3	26.2	22.7	20.2	19.8	17.9	19.1	21.0	19.8	13.6
		MCWBR	12.1	13.0	16.3	17.2	12.0	11.4	10.9	9.2	11.5	14.8	14.0	11.9
(40) Clear-Sky Solar Irradiance	taub		0.255	0.312	0.339	0.363	0.402	0.400	0.388	0.392	0.366	0.321	0.296	0.265
	taud		2.401	2.220	2.316	2.335	2.269	2.301	2.349	2.332	2.401	2.525	2.511	2.446
	Ebn at Noon		279	274	283	285	275	276	278	272	271	272	261	263
	Edh at Noon		23	33	35	38	42	41	38	37	32	25	21	20
	RadAvg		511	856	1229	1448	1734	1881	1936	1647	1301	818	513	372
(45) All-Sky Solar Radiation	RadStd		54	81	127	147	140	139	94	109	96	83	49	38

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page